

# HR Employee Attrition Analysis

SQL / PostgreSQL Project Documentation

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**Project Type:** Data Analytics / HR Analytics

**Primary Tool:** PostgreSQL (pgAdmin)

## 1. Project Overview

This project analyzes employee attrition and workforce patterns using PostgreSQL. The analysis examines employee turnover across departments, salary, tenure, engagement, diversity, compensation, job levels, and bonus percentages.

## 2. Business Objective

The objective is to identify workforce patterns associated with employee attrition and provide data-driven insights that can help understand employee turnover.

Key business questions include:

- Which departments have higher attrition?
- How does salary relate to employee attrition?
- Does employee tenure affect attrition?
- What are the employee engagement patterns?
- How does attrition vary across gender, race/ethnicity, and veteran status?
- How do job levels, compensation, and bonus percentages relate to attrition?

## 3. Datasets Used

The project uses multiple related HR datasets connected primarily through `employee_id` and, for job-profile analysis, `job_title`.

| Dataset             | Purpose                                              |
|---------------------|------------------------------------------------------|
| company_data        | Employee information and attrition analysis          |
| engagement_survey   | Employee engagement analysis                         |
| diversity           | Gender, race/ethnicity, and veteran analysis         |
| job_profile_mapping | Job profile, compensation, level, and bonus analysis |

## 4. Tools & Technologies

- PostgreSQL
- pgAdmin
- SQL

- GitHub for project versioning and presentation

## 5. Data Preparation

Employee IDs were converted to integer format and used as the primary key in `company_data`. Date fields such as `start_date` and `termination_date` were converted from text to DATE format using PostgreSQL date-conversion logic with NULL handling.

The `active_status` field was interpreted as 0 = employee left/terminated and 1 = active employee. The job profile dataset was joined to employee data using `job_title`.

## 6. Analysis Performed

### 6.1 Company Data

- Department-wise attrition
- Salary-wise attrition
- Tenure-wise attrition

### 6.2 Engagement Survey

- Overall engagement score distribution
- Low, medium, and high engagement groups
- Department-wise engagement analysis

### 6.3 Diversity

- Gender distribution and gender-wise attrition
- Race/ethnicity analysis
- Veteran status analysis

### 6.4 Job Profile

- Job level vs attrition
- Compensation level vs attrition
- Bonus percentage vs attrition
- Department and job-profile analysis

## 7. Key Findings

The following findings are based on the completed SQL analysis performed for this project.

- Gender-wise attrition rates were very similar, with only a small difference between male and female employees.
- Employee engagement was analyzed across 2,827 matching survey records.
- Engagement distribution was approximately 34.63% low, 28.30% medium, and 37.07% high.
- The matching engagement survey records contained active employees only, so the survey data does not provide a direct terminated-vs-active engagement comparison.
- Bonus percentage groups showed differences in attrition rates; the 50% bonus group had only 5 employees, so that result should not be treated as strong evidence.
- Compensation, job level, department, tenure, diversity, and engagement were analyzed as separate workforce dimensions.

## 8. Business Insights

The analysis demonstrates how SQL can be used to break down employee attrition into multiple business dimensions instead of relying only on an overall attrition rate. Department, salary, tenure, compensation, job level, bonus structure, engagement, and workforce diversity can each provide different perspectives on employee turnover.

Small sample sizes should be considered before making business decisions. In particular, very small bonus groups can produce unstable attrition percentages and should be interpreted cautiously.

## 9. SQL Concepts Used

- SELECT, WHERE, GROUP BY, HAVING, ORDER BY
- Aggregate functions such as COUNT, AVG, SUM, and MAX
- CASE WHEN for business categorization
- INNER JOIN and multi-table analysis
- Subqueries and correlated subqueries
- Window functions
- Date functions and data-type conversion
- NULL handling and COALESCE

## 10. Project Structure

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■■■■ README.md

■■■■ SQL\_SCRIPTS/

■ ■■■■ COMPANY\_DATA/

■ ■■■■ ENGAGEMENT\_SURVEY/

■ ■■■■ DIVERSITY/

■ ■■■■ JOB\_PROFILE/

■■■■ Screenshots/

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## 11. Conclusion

This project provides practical experience in PostgreSQL-based HR analytics. It demonstrates the ability to clean and prepare relational data, join multiple datasets, write analytical SQL queries, and translate query results into business-oriented workforce insights.

The project can later be extended with Python for exploratory data analysis and Power BI for interactive dashboarding, while keeping the PostgreSQL analysis as the foundation.

Prepared for portfolio / GitHub documentation